

Sajeevan Veeriah

Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer

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Professional summary

Mechatronics engineer combining industrial controls, automotive validation and manufacturing experience with robotics, embedded systems and client software delivery. Connects hardware, control logic and operational data, investigates faults across interfaces, and produces clear verification and handover records. Project work includes ROS 2 autonomy, a 100 Hz sensing prototype and business software developed from requirements through release packaging.

Technical skills

Controls and integration: Programmable logic controllers (PLC), Siemens TIA Portal, HMI/SCADA, iFIX, PVI+, drives, MES, Modbus, PROFINET, industrial IT/OT and GMP documentation.

Robotics and embedded: ROS 2, Nav2, SLAM, EKF sensor fusion, Gazebo, RViz, C/C++, ESP32, STM32, FreeRTOS, CAN, UART, I2C, SPI and BLE.

Software, AI and data: Python, MATLAB/Simulink, TypeScript, React, SQL, Git, CI/CD, OpenCV, scikit-learn and local language-model retrieval workflows.

Design, IoT and test: SolidWorks, Fusion 360, Altium, PCB/enclosure design, 3D printing, Linux, LoRaWAN, MQTT, InfluxDB, Grafana, CANoe and CANalyzer.

Professional experience

Automation and Controls Engineer

Process automation systems integrator | Jan 2026 - Jun 2026

- Integrated PLC logic, HMI/SCADA, field devices, drives and production-data interfaces for pharmaceutical, biotechnology and food manufacturing.
- Migrated validated iFIX content to PVI+ and checked functional behaviour against the existing system. Prepared test and handover evidence under Good Manufacturing Practice (GMP) controls.
- Supported factory and site acceptance testing (FAT/SAT), commissioning and MiR autonomous mobile-robot fault tracing across control, network and production interfaces.

Product Development Test Engineer (Contract)

Global automotive OEM via engineering consultancy | Oct 2025 - Jan 2026

- Tested vehicle-software integration and advanced driver assistance systems (ADAS) through feature-vehicle, breadboard and regression testing.
- Instrumented vehicles, conducted structured test drives and analysed CAN/CAN FD data with Vector CANoe and CANalyzer to isolate faults and document software-readiness evidence.

Technical Officer & Quality Assurance

ABMARC | Jul 2024 - Aug 2025

- Conducted emissions, EV/PHEV range and vehicle-system tests with calibrated instrumentation and data acquisition. Analysed deviations and maintained traceable test, safety and quality records for certification review.

Consultant Engineer, IoT and Projects Administrator

DuxTel | Feb 2024 - Aug 2024

- Built sensor-to-dashboard telemetry using ESP32, LoRaWAN AU915, ChirpStack, MQTT, InfluxDB and Grafana.
- Integrated custom PCB work, CAN/GPS capture, MikroTik connectivity and Linux services for agricultural-equipment monitoring; supported provisioning, field trials and handover.



Manufacturing and quality experience

Production Line Worker / Supervisor / Commissioning Support; Cellar Hand

IDL Australia | Aug 2022 - Feb 2024

- Production roles, Aug 2022 - Sep 2023: led shift output, changeovers and handovers; maintained quality and traceability, recovered faults and supported WestRock/Fibre-King canning-line commissioning checks.
- Cellar Hand, Sep 2023 - Feb 2024: supported sanitation, sampling, quality monitoring and production records.

Production Line Operator and Commissioning Support

Carbon Revolution | Sep 2021 - Dec 2021

- Operated carbon-fibre wheel manufacturing equipment, recorded in-process quality and traceability, and supported equipment trials, setup and first-level fault recovery.

Undergraduate Quality Engineer

Thornton Engineering Australia | Mar 2018 - Mar 2020

- Supported drawing review, inspection and test plans (ITPs), manufacturing data reports (MDRs), material traceability and AS/NZS compliance in structural-steel fabrication.

Selected engineering projects

Autonomous navigation rover | ROS 2

Built a differential-drive platform with LiDAR and IMU sensing. Integrated SLAM, EKF localisation and Nav2 planning in C++/Python; checked transforms, paths and recovery in Gazebo and RViz. Hardware build with autonomy validated in simulation.

Ataxia assessment device | Embedded mechatronics

Designed an ESP32 prototype, Altium PCB and enclosure with four Hall-effect channels and 100 Hz acquisition. Added Bluetooth display and reporting; used MATLAB to test accuracy, reversals, drift and temperature behaviour.

SWL pricing and inventory control | Client software

Sole engineer across requirements, pricing rules, React/TypeScript, Tauri/Rust, SQLite, automated tests and CI. Built operator-reviewed ServiceM8 imports with change, exception, rollback and audit reports; completed release packaging.

VeerAI | Local AI system

Built a locally deployed system around an open-weight small language model with document ingestion, retrieval, memory, tools and evaluation.

Community platforms and engineering tools

Built the Newcomb and District Cricket Club platform and Engineering Mastery Lab. Combined user workflows, engineering calculations, technical content and software testing.

Education and additional information

Bachelor of Mechatronics Engineering (Honours), Distinction

Deakin University | 2025

Higher National Diploma in Mechatronics, Robotics and Automation Engineering

Cardiff Metropolitan University | Distinction | 2016

Member, Engineers Australia. English, Tamil and Sinhala. Community leadership and engineering mentoring.

Full Australian work rights: subclass 485, valid until 20 March 2028. Available immediately; willing to relocate.